

Structured Training Program: Wi-Fi

Module 1: Introduction and History of Wi-Fi

1. In which OSI layer the Wi-Fi standards/protocol fits?

The Wi-Fi and its protocol standards fits in the OSI layer 1 – Physical Layer and OSI layer 2 – Datalink Layer.

Reasons for Layer 1:

- Signal Transmission: Wi-Fi operates using radio waves (2.4 GHz, 5 GHz, 6 GHz bands) to transmit raw bits between devices.
- Physical Specifications: 802.11 standards define the physical specifications, including modulation techniques (like OFDM, DSSS) and transceiver control.
- Hardware Interface: It manages the network interface card (NIC) and antenna interaction to put data onto the wireless medium

Reasons for Layer 2:

- Framing: Wi-Fi organizes bits received from the Physical layer into structured frames (802.11 frames).
- Media Access Control (MAC): Wi-Fi uses a MAC sublayer (IEEE 802.11 MAC) to manage how devices access the shared airwaves, utilizing techniques like Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA).
- Node-to-Node Addressing: It utilizes MAC addresses to ensure frames are delivered to the correct device within the local network segment.
- Error Detection and Flow Control: It provides mechanisms for checking data errors and controlling data flow to ensure reliable transmission between nodes.

2. Can you share the Wi-Fi devices that you are using day to day life share that device's wireless capability / properties after connecting to the network. Match your device to Wi-Fi generation based on the capability

Device Name	Freq Band	Max Bandwidth / Channel	Non-Overlapping Channels	Max MIMO	Highest Modulation	Max PHY Data Rate	MU-MIMO	MU-OFDMA (Shared Bandwidth)	TWT	Wi-Fi Generation
OnePlus Nord CE3 Lite 5G	2.4 GHz, 5 GHz	80 MHz	(approx) 6 (5 GHz)	2x2	1024-QAM	(approx) 1200 Mbps	Yes	No	No	Wi-Fi 5 (802.11ac)
Cisco Wi-Fi 6 Router	2.4 GHz, 5 GHz	160 MHz	(approx) 8+	4x4	1024-QAM	(approx) 4800 Mbps	Yes	Yes	Yes	Wi-Fi 6 (802.11ax)
Dell Inspiron 14 (2024)	2.4 GHz, 5 GHz	160 MHz	(approx) 8+	2x2	1024-QAM	(approx) 2400 Mbps	Yes	Yes	Yes	Wi-Fi 6
Redmi Pad 6	2.4 GHz, 5 GHz	80 MHz	(approx) 6	2x2	1024-QAM	(approx) 1200 Mbps	Yes	No	No	Wi-Fi 5 (802.11ac)
TP-Link Home Router	2.4 GHz, 5 GHz	80–160 MHz	(approx) 6–8	2x2 or 4x4	1024-QAM	(approx) 1200–3000 Mbps	Yes	Partial	No/Yes	Wi-Fi 5 / Wi-Fi 6

3. What is BSS and ESS ?

Basic Service Set (BSS):

Basic Service Set (BSS) is a single Access Point (AP) connecting devices within a small range (called as cell) through a wireless medium commonly used in home routers.

Extended Service Set (ESS):

Extended Service Set (ESS) connect more than one BSS usually through Ethernet and allows user within the ESS to roam freely under a single Service Set Identifier (SSID).

Criteria	BSS	ESS
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Structure	Consist of one AP and Client	Consist of 2 or more BSSs (multiple APs)
Coverage	Small (one cell)	Large (multiple cells)
SSID	Identified by MAC address of AP (BSSID)	All BSSs in ESS share the same service set identifier (SSID) and are distinguished by BSSIDs of individual AP
Roaming	Not possible without disconnection	Allows user to move between APs without disconnection
Example	A home router	University Campus with many APs

4. What are the basic functions of Wi-Fi access points ?

Basic Functions of Wi-Fi Access Points:

- **Bridging Wired to Wireless:** Connects directly to a wired Ethernet router, switch, or hub and converts that data into a wireless signal, allowing wireless devices to access the wired network.
- **Extending Coverage (Repeater Mode):** Acts as a repeater by extending the Wi-Fi signal to areas with low signal strength or dead spots, such as in large offices or homes.
- **Wireless Connectivity Hub:** Acts as the central point for wireless communications, facilitating communication between connected clients.
- **VLAN Support (Segmentation):** Supports multiple SSIDs, enabling segmentation of the network to provide separate connections for guests, IoT devices, or employees.

- Bridging Different Network Types: Can be configured to connect wireless-only devices to wired infrastructure, acting as a workgroup bridge.

5. Difference between Bridge Mode and Repeater Mode.\

Bridge Mode	Repeater Mode
Connect two different routers	Extend the network coverage to dead zones
Maintain faster Speed	Reduce throughput as they act as Secondary APs
Best for connecting Distant Wireless Devices that need stable connection without running cable	Best for boosting signals into dead zones for mobile devices
Entire bandwidth is used for Secondary AP	Bandwidth is not fully used for Secondary AP they are split to use across both Primary and secondary AP equally using the same Wireless channel

6. What are the difference between IEEE 802.11a and IEEE 802.11b

Feature	IEEE 802.11a	IEEE 802.11b
Frequency	5 GHz	2.4 GHz
Max Speed	54 Mbps	11 Mbps
Range	Shorter	Longer
Interference	Low	High

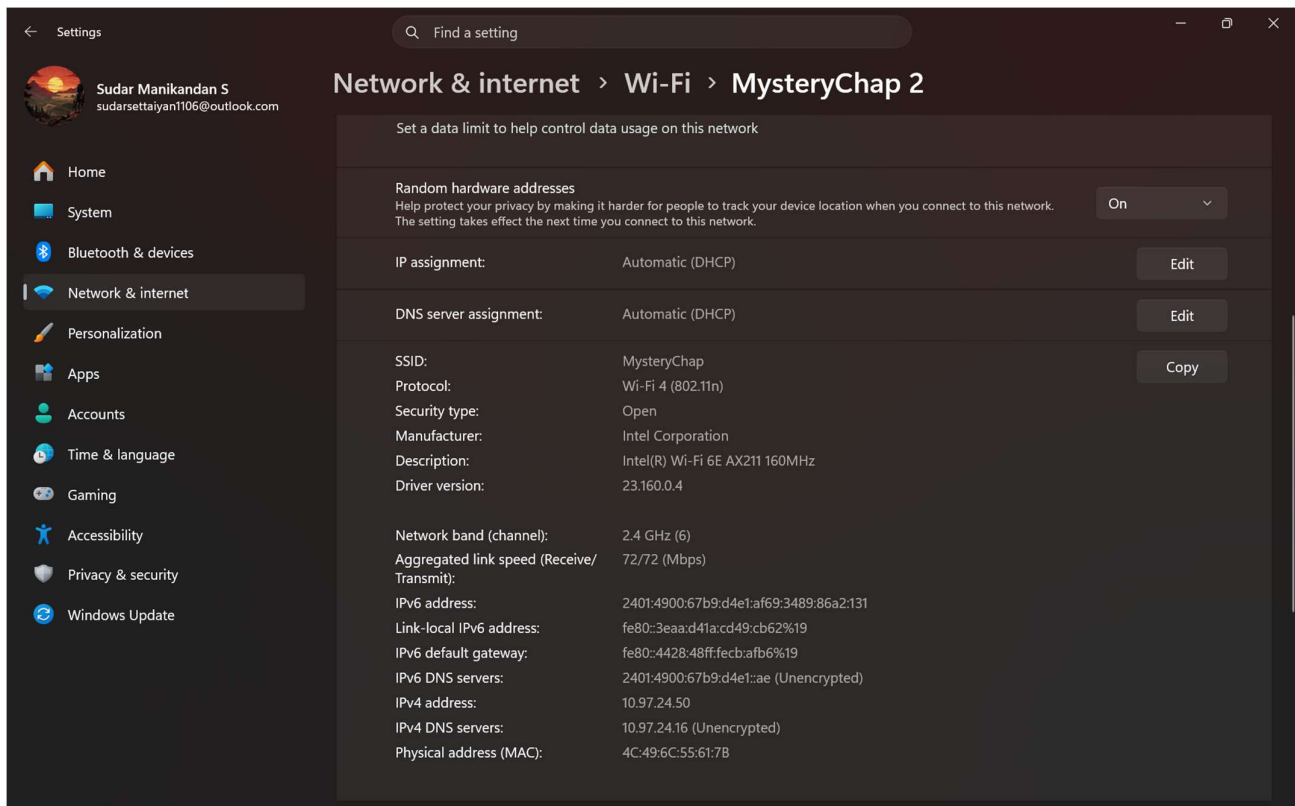
Modulation	OFDM	DSSS
Channels	More	Limited (3)
Cost (initially)	Higher	Lower

7. Configure your modem/hotspot to operate only in 2.4 Ghz and connect your laptop / WI-Fi device and capture the propertoies in your wifi device. Repeat the same in 5Ghz and tabulate the difference you observe during this.

Properties in 2.4Ghz

Device used for hotspot: Oneplus Nord CE3 Lite 5G (Intel(R) Wi-Fi 6E AX211 160MHz)

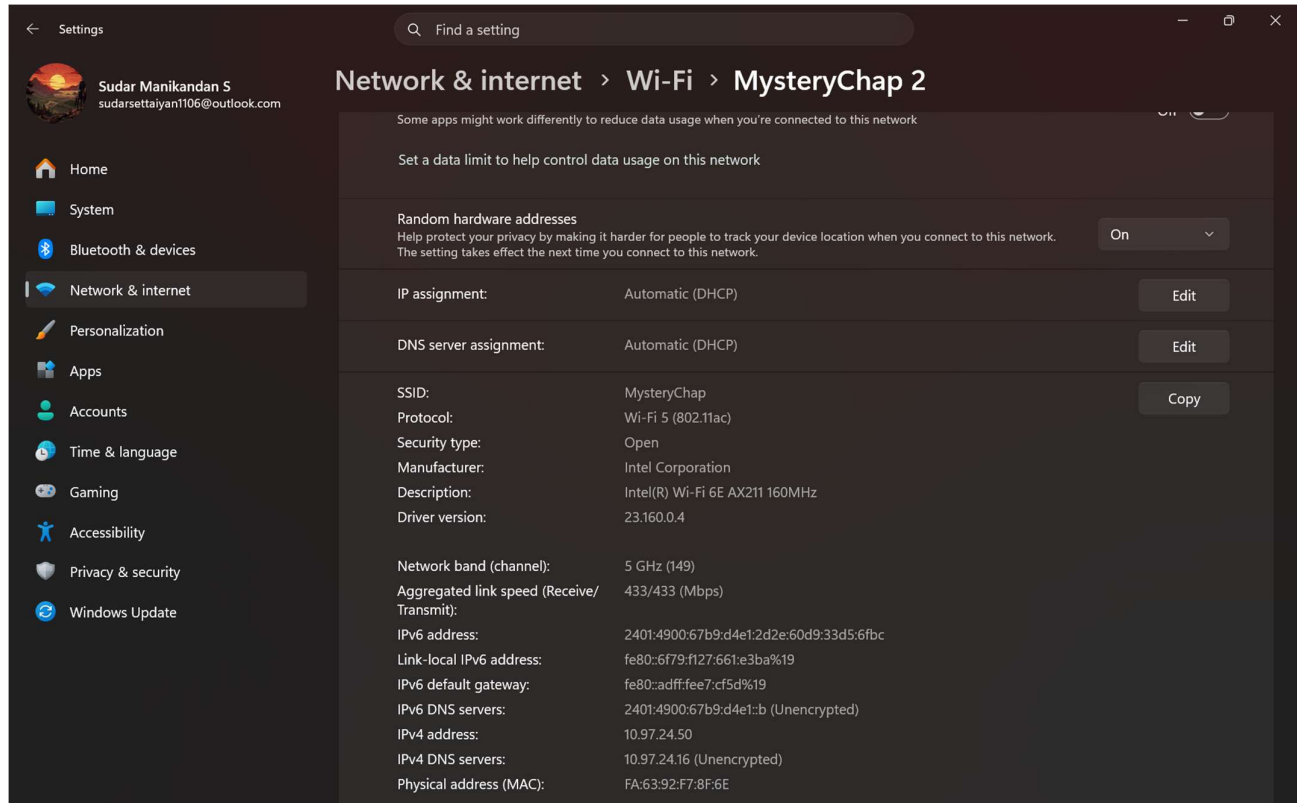
Device used for Connection : Dell Inspiron 14 (Wi-Fi card - Intel(R) Wi-Fi 6E AX211 160MHz)



Properties in 5 GHz:

Device used for hotspot: Oneplus Nord CE3 Lite 5G (Intel(R) Wi-Fi 6E AX211 160MHz)

Device used for Connection : Dell Inspiron 14 (Wi-Fi card - Intel(R) Wi-Fi 6E AX211 160MHz)



Feature	2.4 GHz	5 GHz
Wi-Fi Standard	Wi-Fi 4 (802.11n)	Wi-Fi 5 (802.11ac)
Frequency Band	2.4 GHz	5 GHz
Channel	6	149
Speed (Rx/Tx)	72 / 72 Mbps	433 / 433 Mbps
Range	Longer	Shorter
Wall Penetration	Better	Weaker
Interference	High (inference from Bluetooth, microwave, etc.)	Low
Network Stability	More stable at distance	Best when near router

8. What is the difference between IEEE and WFA ?

Basis of Comparison	IEEE	WFA (Wi-Fi Alliance)
Full Form	Institute of Electrical and Electronics Engineers	Wi-Fi Alliance
Nature	Professional standards organization	Industry consortium
Primary Role	Develops technical standards	Certifies Wi-Fi devices
Main Function	Defines protocols (e.g., 802.11 series)	Ensures interoperability between devices
Focus Area	Design and specification of wireless communication	Testing, compatibility, and user experience
Output	Standards documents (802.11a/b/g/n/ac/ax)	"Wi-Fi CERTIFIED" label
Target Users	Engineers, researchers, developers	Manufacturers and consumers
Example	Defines how Wi-Fi works (e.g., modulation, frequency)	Tests if router and mobile can work together
Responsibility	Standard creation	Product validation
Dependency	Independent body	Uses IEEE standards for certification

9. List down the types of Wi-Fi internet connectivity backhauls share your home/college's wireless internet connectivity backhaul name and its properties.

Types of Wi-Fi connectivity Backhauls:

- a) Fibre Optics
- b) MoCA
- c) Satellite
- d) Cellular
- e) Powerline Backhaul

The type of backhaul used in our college is Fibre Optics Backhaul:

Properties of Fibre Optics Backhaul:

- Low Latency

- Immune to interference
- Capable of transferring High bandwidth over long distance
- High reliability and security

The type of backhaul used in our home is Cellular Backhaul:

Properties of Cellular Backhaul:

- Capable of handling large volume of data from multiple user (through 4G/LTE and 5g technology)
- Rapid deployment and easy mobility
- Variable Latency unlike Fibre Optics

10. List down the Wi-Fi topology and their use cases.

Wi-Fi topologies and their use cases:

1. Infrastructure Mode

Description: Devices connect through a central Access Point (AP) (router).

Use cases:

- Home Wi-Fi networks
- College/office networks
- Internet access for multiple devices

2. Repeater Mode

Description: Extends existing Wi-Fi by receiving and re-transmitting signals.

Use cases:

- Eliminating dead zones in large homes
- Extending campus/building coverage
- Improving weak signal areas

3. Bridge Mode

Description: Connects two networks (wired or wireless) at Layer 2.

Use cases:

- Connecting two buildings wirelessly
- Linking LANs without running cables
- Backbone connections in campuses

4. Ad-Hoc Mode

Description: Direct device-to-device connection without an AP.

Use cases:

- Quick file sharing between laptops
- Temporary networks (meetings, labs)
- Peer-to-peer communication

5. Mobile Hotspot Mode

Description: Smartphone or device acts as a Wi-Fi router.

Use cases:

- Sharing mobile data
- Internet access during travel
- Backup connection when broadband fails

6. Mesh Mode

Description: Multiple nodes form a self-healing network with seamless roaming.

Use cases:

- Smart homes with large coverage
- Hotels, malls, campuses
- IoT-heavy environments

7. Workgroup Bridge Mode (WGB)

Description: Connects wired devices to a wireless network via a bridge.

Use cases:

- Connecting printers, CCTV, or legacy devices
- Industrial equipment networking
- Smart classrooms/labs

8. IoT Gateway Mode

Description: Acts as a bridge between IoT devices and cloud/network.

Use cases:

- Smart home automation
- Industrial IoT (IIoT)
- Sensor data collection and processing